

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY

SOFTWARE ENGINEERING

PROJECT DOCUMENTATION

AI-Based Stock Market Analysis and Prediction System

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1. Project Overview

The AI-Based Stock Market Analysis and Prediction System aims to develop an intelligent stock market analysis platform that helps users understand stock behaviour, analyze market trends, and simulate investment decisions using Artificial Intelligence and Machine Learning techniques. The system processes historical market data, extracts important patterns, predicts future stock trends, and provides meaningful insights through an interactive platform.

2. Problem Statement

Stock market analysis requires users to manually study large volumes of financial data, technical indicators, and market movements. This makes it challenging for beginners to understand stock behaviour and practice investment strategies. This project addresses this challenge by providing an AI-based analysis system along with a virtual investment simulation environment.

3. Proposed System

Module	Description
Data Processing	Historical data collection, preprocessing, and preparation.
AI/ML Analysis	Machine learning based prediction and trend analysis.
Portfolio Simulation	Virtual investment, buy/sell operations, and performance tracking.
Visualization	Displaying stock trends, predictions, and portfolio information.

4. System Workflow

Historical Market Data → Data Preprocessing → Feature Engineering → Machine Learning Model → Prediction & Trend Analysis → Visualization and Portfolio Simulation

5. Machine Learning Approach

The system follows an AI-centric approach where historical stock data is transformed into meaningful features. Technical indicators such as Moving Average, Daily Return, Volatility, and Volume-related features are used by machine learning models to identify market patterns and generate predictions.

6. Key Features

- Historical stock data analysis
- Machine learning based stock prediction
- Trend identification (Bullish, Neutral, Bearish)
- Virtual portfolio simulation using fake capital
- Buy/Sell tracking and portfolio performance analysis
- Stock insights and visualization

7. Current Implementation Status

Completed components include historical dataset integration, data preprocessing, feature engineering, machine learning model training, prediction prototype, and model evaluation.

8. Future Enhancements

- Live market data integration
- Improved directional prediction
- Advanced recommendation engine
- Real-time dashboard
- Enhanced simulation experience

Conclusion

The proposed system combines software engineering principles with artificial intelligence to create an intelligent stock market analysis and learning platform. It aims to make market understanding easier while providing users with a safe environment to explore investment strategies.